

Introduction to Deep Learning

Session 9: Convolutional Neural Networks

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Lecture Overview

1. Why Not MLPs for Images?

2. Convolution Operation

3. Pooling

4. CNN Architecture

Images in Computers

Image: 3D tensor of shape $C \times H \times W$

- Each entry: pixel intensity, integer 0–255 or float 0–1,
- 2^8 possible values ~ 8 bits
- **Grayscale** ($C = 1$): single channel, 0 = black, 255 = white
- **RGB** ($C = 3$): red, green, blue — standard cameras, screens

Other formats:

- **RGBA** ($C = 4$): RGB + α transparency
- **CMYK** ($C = 4$): print format
- **Medical/hyperspectral**: many channels, often 16-bit depth

DL mostly **RGB and grayscale** — other formats typically converted before use

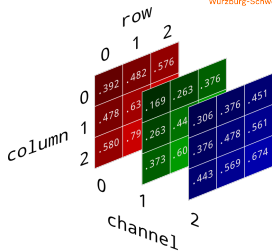


Image sizes — from benchmarks to real world:

Source	Size	Channels	Pixels
MNIST	28×28	1	784
ImageNet	224×224	3	150 528
Phone camera	4000×3000	3	36 000 000
Full HD screen	1920×1080	3	6 220 800
Medical (CT slice)	512×512	1	262 144

MLP approach: flatten image to vector $\mathbf{x} \in \mathbb{R}^{H \cdot W \cdot C}$, apply fully connected layers

ImageNet: first hidden from $\mathbb{R}^{150\,528} \rightarrow \mathbb{R}^{1024} \Rightarrow \approx 154\text{M}$ parameters — one layer only!

How to handle high-dimensional data efficiently?

How to Look at Images

Human vision:

- Looks for local patterns — edges, corners, textures
- Same pattern recognised anywhere in the visual field
- Simple features combine into complex ones

Technical equivalent:

- **Local connectivity** — connect to small patches only
- **Parameter sharing** — same filter at every location
- **Hierarchical features** — edges → shapes → objects

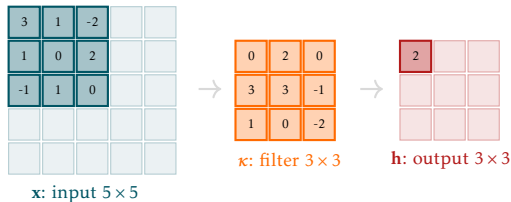
Three key properties: **locality, translation invariance, parameter efficiency**

These define the **convolutional layer**

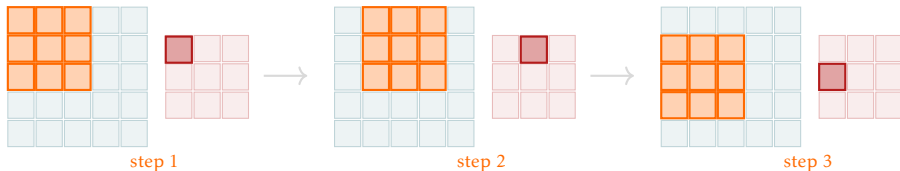
Convolution Operation

Convolution — The Operation

Single step: dot product of local patch and filter $\rightarrow$ one output value



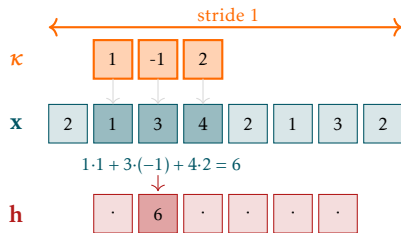
Full operation: same filter slides across input — each position produces one output value



All 3 requirements: local connectivity, parameter sharing, translation invariance

1D Convolution

Filter $\kappa \in \mathbb{R}^k$ slides over signal $\mathbf{x} \in \mathbb{R}^n$ — at each position i , dot product with local patch



$$(\mathbf{x} * \kappa)[i] = \sum_{j=0}^{k-1} \mathbf{x}[i+j] \kappa[j]$$

- i — output position; j — index within filter
- Dot product of filter with local patch

Example: at position $i = 1$:

$$1 \cdot 1 + 3 \cdot (-1) + 4 \cdot 2 = 6$$

2D Convolution

Extend to 2D: filter $\kappa \in \mathbb{R}^{k \times k}$ slides over image $\mathbf{X} \in \mathbb{R}^{H \times W}$:

$$(\mathbf{X} * \kappa)[i, j] = \sum_{p=0}^{k-1} \sum_{q=0}^{k-1} \mathbf{X}[i + p, j + q] \kappa[p, q]$$

3	1	-2		
1	0	2		
-1	1	0		

$\mathbf{x}$: input 5×5



0	2	0
3	3	-1
1	0	-2

κ : filter 3×3



2		

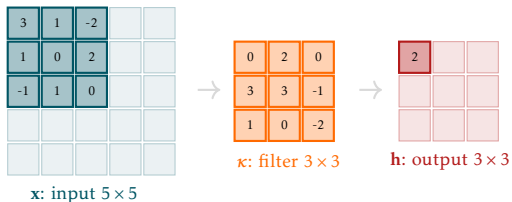
$\mathbf{h}$: output 3×3

- (i, j) — output position
- (p, q) — index within filter
- Two sums: over rows and columns of filter

Output size: $(H - k + 1) \times (W - k + 1)$

Convolution as Matrix Multiplication

Flatten input and output — convolution becomes matrix multiplication

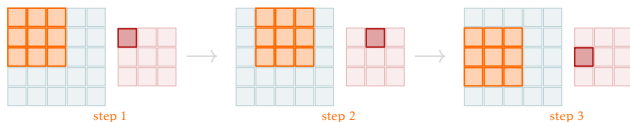


Convolution = FC layer with **sparse** and **tied** weights

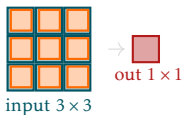
Convolution - **linear transformation!**

Padding

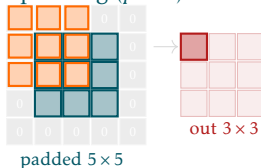
Problem: convolution shrinks spatial dimensions — each layer loses $k - 1$ pixels



no padding



same padding ($p = 1$)



Valid padding ($p = 0$)

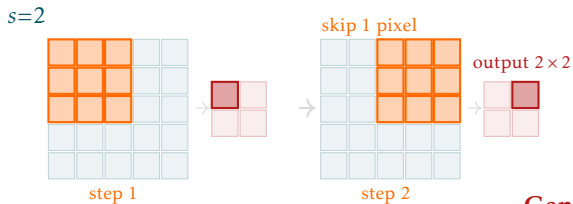
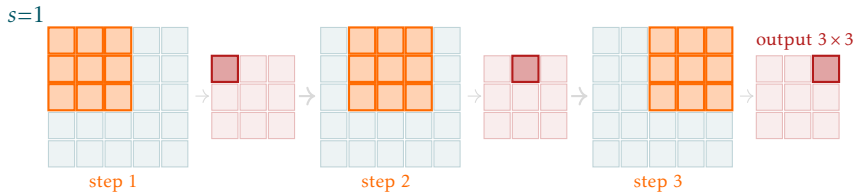
- No zeros added
- Output shrinks: $(H - k + 1) \times (W - k + 1)$
- Spatial information lost at borders

Same padding ($p = (k - 1)/2$)

- Zeros added around border
- Output same size as input
- Standard choice in most architectures

Stride

Stride s : filter κ moves s pixels at each step instead of 1



Stride 1: κ moves one step a time

Stride 2: κ skips one - output halved

General output size: $\left\lceil \frac{H + 2p - k}{s} \right\rceil + 1$

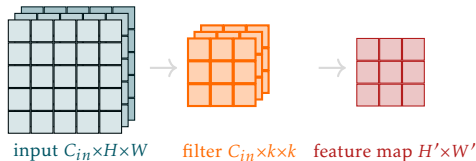
Multiple Input Channels

RGB input: C_{in} channels — filter must have same depth C_{in}

Operation: filter has C_{in} slices — each slice convolves with one input channel

$$(\mathbf{X} * \kappa)[i, j] = \underbrace{\sum_{c=0}^{C_{in}-1} \sum_{p=0}^{k-1} \sum_{q=0}^{k-1} \mathbf{X}[c, i+p, j+q] \kappa[c, p, q]}_{\text{conv with channel } c}$$

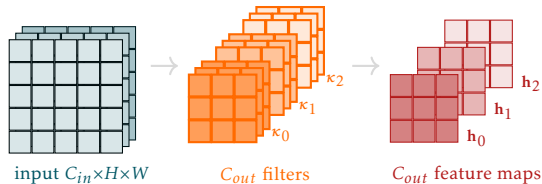
- Inner double sum: convolution of filter slice c with input channel c
- Outer sum over c : results summed across all channels
- One filter $\rightarrow$ one output feature map



Multiple Output Channels

C_{out} **filters**: each filter independently applied to the full input volume

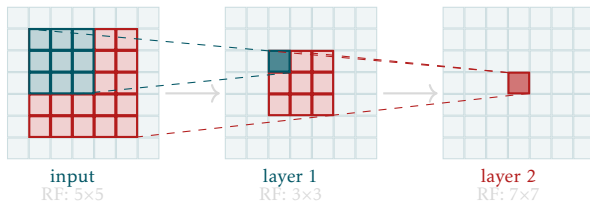
$$\mathbf{h}_f[i, j] = \sum_{c=0}^{C_{in}-1} \sum_{p=0}^{k-1} \sum_{q=0}^{k-1} \mathbf{X}[c, i+p, j+q] \kappa_f[c, p, q] \quad f = 0, \dots, C_{out} - 1$$



- Each filter κ_f produces one feature map $\mathbf{h}_f$ independently
- Output tensor: $C_{out} \times H' \times W'$
- Parameters: $C_{out} \times C_{in} \times k \times k + C_{out}$ (with bias)

Receptive Field

Receptive field: region of the input that influences a given neuron's output



- Each 3×3 conv layer adds 2 pixels to the receptive field in each dimension
- L layers of 3×3 filters: receptive field = $(2L + 1) \times (2L + 1)$
- Deep networks see large regions despite small local filters
- Early layers: local features (edges) — later layers: global patterns (objects)

Pooling

Pooling: spatial downsampling — reduces feature map size

Max pooling:

$$y[i, j] = \max_{p, q \in \mathcal{R}_{ij}} x[p, q]$$

- Takes maximum over local region
- Retains strongest activation
- Most commonly used

Average pooling:

$$y[i, j] = \frac{1}{|\mathcal{R}_{ij}|} \sum_{p, q \in \mathcal{R}_{ij}} x[p, q]$$

- Takes average over local region
- Smoother downsampling
- Less common in hidden layers

Benefits:

- Reduces spatial dimensions — fewer parameters in subsequent layers
- Local translation invariance — small shifts in feature position do not affect output

Global average pooling (GAP): averages entire feature map to a single value per channel

$$y_c = \frac{1}{H \cdot W} \sum_{i=1}^H \sum_{j=1}^W x_c[i, j]$$

- Output: vector of length C_{out} — one value per feature map
- Replaces large FC layers at the end of the network
- Dramatically reduces parameters
- Forces each feature map to represent a single concept

Used in: ResNet, GoogLeNet, MobileNet — modern standard

In PyTorch: `nn.AdaptiveAvgPool2d((1, 1))` followed by `flatten`

CNN Architecture

Standard pattern: Conv $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ (Pool)

- **Conv:** extract features via learned filters
- **BatchNorm:** stabilise activations — essential for deep CNNs
- **ReLU:** non-linearity — enables complex feature combinations
- **Pool:** spatial downsampling — reduce resolution, increase receptive field